

33 — SIPO Capstone: Constraint-Conditioned Addressed Update Law

Professional reading edition of the evolving public MKUFT canon

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1. Why this capstone exists

Layer Before Law introduced

$$\mathcal{U}_t \xrightarrow{\text{Update}_{\text{SIPo}}} \mathcal{U}_{t+\Delta}$$

as a deliberately undefined placeholder. The later canon supplied the missing dependencies: addressed state-space families, load-bearing relation sets, completion fibers, macro realisation classes, intrinsic/extrinsic address, temporal transport, context-indexed P-realisation, and an explicit P-only null for candidate I→P effects.

The remaining task is therefore not to invent another phenomenon. It is to state the smallest update object that composes those dependencies without flattening their mathematical types.

The core result is:

the admissible domain on which an effective law acts is part of the law object, but it is not necessarily the only relationally conditioned part. A complete candidate update must distinguish domain/support, transition rule, and weighting, then state which of those objects is changed by the addressed relational state.

This is the operational content of **Layer Before Law**.

2. Ambient S versus effective possibility

Retain the substrate scaffold

$$S^* = (\Omega^*, \Sigma^*, \mu^*)$$

as the ambient structured-possibility hypothesis.

Do not silently interpret every recursive update as changing the whole ambient substrate. At time t , use an effective possibility object

$$S_t^{\text{eff}} = (\Omega_t^{\text{eff}}, \Sigma_t^{\text{eff}}, \mu_t^{\text{eff}}), \quad \Omega_t^{\text{eff}} \subseteq \Omega^*,$$

where the active support, sigma-algebra and measure are declared by the implementation.

The conservative meaning of the recursive 0→S leg is:

a realised physical/registered outcome changes the state, history, context and relations from which the next effective possibility set is computed.

That does not imply that observation rewrites a universal substrate.

3. The physical law object

When domain or boundary conditions are load-bearing, a differential expression, transition matrix, Hamiltonian symbol or field equation is not a complete dynamical specification by itself.

Define a typed physical law object

$$\mathcal{L}_{P,t} = (\mathcal{D}_{P,t}, \mathcal{T}_{P,t}, \mathcal{W}_{P,t}).$$

- $\mathcal{D}_{P,t}$ — addressed physical state/path/operator domain on which the law is valid;
- $\mathcal{T}_{P,t}$ — lawful transition relation, generator, propagator, flow, field equation or solution relation;
- $\mathcal{W}_{P,t}$ — optional domain-appropriate weighting object, such as a deterministic choice rule, probability kernel, quantum amplitude/channel structure, or another explicitly typed weighting.

No universal mathematical type is imposed on $\mathcal{W}_{P,t}$. Classical probabilities and quantum amplitudes are not merged merely to keep one notation.

The law object therefore carries both the rule **and the conditions under which that rule is the law of the addressed object being described**.

4. Full addressed admissibility descriptor

Let the current complete addressed state be

$$\mathcal{U}_t = (S_t^{\text{eff}}, I_t, P_t, O_t, b_t, H_t),$$

with measured/controlled environment E_t supplied where required by the claim.

Let

$$\Xi_t = \mathcal{Q}_t(\mathcal{U}_t, E_t)$$

be the **typed admissibility descriptor** produced by the relation/address/compatibility architecture inherited from Module 32S3 and refined by 32S4. Ξ_t is not required to be one scalar. It may contain typed constraints, support restrictions, boundary/context data, relation scopes, path/history restrictions, or other objects required to construct the active P-law.

Including the current P-state explicitly prevents the admissibility construction from being treated as detached from the physical state it constrains. Including O_t does not grant observer magic: an O-variable earns a physical role only if operationally defined and if any claimed independent effect survives the strongest adequate P-state/history/measurement-context null.

5. Law assembly — general capstone dependency

Define a **physical-law assembler**

$$\mathfrak{C}_P : (\Xi_t, P_t, E_t) \mapsto \mathfrak{L}_{P,t} = (\mathcal{D}_{P,t}, \mathcal{T}_{P,t}, \mathcal{W}_{P,t}).$$

This is an architectural typing map, not a claim that one universal closed-form function already generates every physical law.

The most conservative and best-controlled branch is **domain-only conditioning**:

$$\mathcal{D}_{P,t} = \mathfrak{D}_P(\Xi_t),$$

while $\mathcal{T}_{P,t}$ and $\mathcal{W}_{P,t}$ are the standard P-law objects restricted or realised on that domain.

But the general capstone must also allow a candidate relation to change, if independently demonstrated:

- the admissible domain/support $\mathcal{D}_{P,t}$;
- the transition rule/generator $\mathcal{T}_{P,t}$;
- the weighting/selection object $\mathcal{W}_{P,t}$;
- or a typed combination.

Thus **domain before generator** remains the minimal clean Layer-Before-Law branch, not an imposed claim that every possible I→P effect must occur only through domain selection.

6. The SIPO update

Let

$$P_t = (\beta_t, p_t) \in \mathfrak{P} = \bigsqcup_{\beta \in A_P} \{\beta\} \times \mathcal{P}_\beta.$$

Construct

$$\begin{aligned} \Xi_t &= \mathcal{Q}_t(\mathcal{U}_t, E_t), \\ \mathfrak{L}_{P,t} &= \mathfrak{C}_P(\Xi_t, P_t, E_t). \end{aligned}$$

Let

$$P_{t+\Delta}^- \in \text{Propagate}_{P,\Delta}^{\mathfrak{L}_{P,t}}(P_t)$$

represent the pre-registration physical result. The propagation object may return a singleton, probability law, evolved quantum state/channel, or solution family according to the declared P-model.

Measurement/registration is typed separately:

$$\text{Instrument}_{P \rightarrow (P,O)} : P_{t+\Delta}^- \mapsto (P_{t+\Delta}, O_{t+\Delta}).$$

An actual measurement interaction can alter P while creating an O-record. Passive observation is the special case in which the instrument leaves the relevant P-state unchanged.

Finally, let $\text{Readdress}_{t+\Delta}$ update typed address, retained history, relation/bracket structure, effective possibility and any promoted higher-scale variables.

The capstone is

$$\text{Update}_{\text{SIPO},\Delta}(\mathcal{U}_t) = \text{Readdress}_{t+\Delta} \left[\mathcal{U}_t, \text{Instrument}_{P \rightarrow (P,O)} \left(\text{Propagate}_{P,\Delta}^{\mathfrak{C}_P(\mathcal{Q}_t(\mathcal{U}_t, E_t), P_t, E_t)}(P_t) \right) \right].$$

The strongest readable operator form is

$$\text{Update}_{\text{SIPO}} = \text{Readdress} \circ \text{Instrument}_{P \rightarrow (P,O)} \circ \text{Propagate}_{P,\Delta}^{\mathfrak{L}_P}, \quad \mathfrak{L}_P = \mathfrak{C}_P \circ \mathcal{Q} \text{ (with typed state/context arguments).}$$

The final composition shorthand suppresses the explicit P, E arguments for readability; the expanded equation above controls.

Plain-language form:

```
current addressed S/I/P/O state + context/history
→ typed admissibility descriptor  $\Xi$ 
→ assemble the active P-law object (domain, transition rule, weighting)
→ propagate physically
→  $P \rightarrow (P,O)$  instrument creates physical measurement back-action + record
→ realised path + record + context readdress the next effective possibility structure
→ repeat
```

7. Readdressing closes the loop

Where a path/history descriptor is defined, let

$$\gamma_t : P_t \rightsquigarrow P_{t+\Delta}$$

represent the realised transition, including the physical measurement interaction where load-bearing.

Update history by

$$H_{t+\Delta} = H_t \oplus (\gamma_t, O_{t+\Delta}),$$

where $\oplus$ is typed append/update, not numerical addition.

Extract the new extrinsic/contextual address via

$$b_{t+\Delta} = B(P_{t+\Delta}, E_{t+\Delta}).$$

Recompute the relation state/bracket through

$$I_{t+\Delta} = R_I(I_t, P_{t+\Delta}, O_{t+\Delta}, b_{t+\Delta}, H_{t+\Delta}).$$

In the physical-only null, R_I may reduce entirely to a function of adequate P-state/history and registered record.

The next effective possibility object is

$$S_{t+\Delta}^{\text{eff}} = R_S(S_t^{\text{eff}}, I_{t+\Delta}, P_{t+\Delta}, O_{t+\Delta}, b_{t+\Delta}, H_{t+\Delta}).$$

The minimum interpretation is recomputation of the next reachable/admissible support and associated measure. A stronger claim that ambient S^* changes is not inherited from the recursion.

Thus

$$\mathcal{U}_{t+\Delta} = (S_{t+\Delta}^{\text{eff}}, I_{t+\Delta}, P_{t+\Delta}, O_{t+\Delta}, b_{t+\Delta}, H_{t+\Delta}).$$

8. Support before weighting

The capstone separates:

1. **which states or paths are admissible;**
2. **what transition rule acts on that domain;** and
3. **how admissible alternatives are selected or weighted**, where a separate weighting is required.

Thus the dependency order is

$$\text{addressed admissibility} \rightarrow \text{domain} \rightarrow \text{domain-valid transition rule} \rightarrow \text{domain-appropriate weighting/selection}.$$

This is a logical typing order, not a universal temporal sequence. In some physical formalisms transition and weighting structure are inseparable components of one law object; the tuple notation preserves that possibility rather than forcing an artificial decomposition.

9. Path, history and scale

When endpoint state is insufficient, use an admissible path family

$$\Gamma_{t,\Delta}^{\text{adm}}(P_t) = \{\gamma : \gamma(t) = P_t, \gamma(s) \in \mathcal{D}_{P,s}, \text{ declared relation/transport constraints pass}\}.$$

A path-dependent propagator may depend on this family rather than endpoint alone. If path history adds no preregistered value beyond ordinary hysteresis, drift or memory, reduce to endpoint-only modelling.

At scale, let

$$y_L = R_{\ell \rightarrow L}(a_\ell),$$

with lower realisation class

$$\mathcal{M}_{y_L} = \mathfrak{C}_{B,\ell} \cap R_{\ell \rightarrow L}^{-1}(y_L).$$

Only after Module 32S promotion tests pass may the effective whole be inserted as a typed constituent of the next-scale bracket. It then participates in the next $\mathcal{Q}$ calculation and can change the next law object $\mathfrak{L}_{P,L}$.

Thus:

lower relations constrain admissibility
 → stable higher object is promoted
 → promoted whole becomes a typed constituent/relation
 → next-scale admissibility changes
 → next effective law object is assembled

10. Deterministic/viability specialisation

For set-valued dynamics

$$\dot{p}_t \in F_P(p_t, t)$$

and admissible set $\mathcal{D}_{P,t}$, a viability-style specialisation is

$$\dot{p}_t \in F_P(p_t, t) \cap T_{\mathcal{D}_{P,t}}(p_t),$$

where $T_{\mathcal{D}}(p)$ is an appropriate contingent/tangent cone when its mathematical hypotheses hold.

If the intersection is empty, do not invent a transition. The state may leave the domain, terminate, trigger a bracket change, or expose an inconsistent model depending on implementation.

11. Stochastic specialisation

For a reference kernel

$$K_P^0(dp' | p)$$

and admissibility function

$$a_t(p, p') \in [0, 1],$$

one **conditioning** specialisation is

$$Z_t(p) = \int a_t(p, q) K_P^0(dq | p),$$

$$K_{P,t}^{\text{adm}}(dp' | p) = \frac{a_t(p, p') K_P^0(dp' | p)}{Z_t(p)}, \quad Z_t(p) > 0.$$

This is not universal: reflecting, absorbing, driven, controlled, non-Markovian and other constrained processes require their correct physical generators rather than post-hoc renormalisation.

For path-weight formulations, one prior-art specialisation is

$$d\mathbb{P}_{\text{adm}}[\gamma] = \frac{\mathbf{1}_{\Gamma^{\text{adm}}}[\gamma] e^{-\Phi(\gamma)}}{Z} d\mathbb{P}_P^0[\gamma],$$

where Φ is dimensionless and independently justified. $\Phi = 0$ gives pure support conditioning.

12. Quantum specialisation

Quantum mechanics gives an exact precedent for domain-before-dynamics: boundary conditions can select self-adjoint extensions and therefore physically admissible unitary evolution.

For

$$H_t : \mathcal{D}(H_t) \subseteq \mathcal{H}_t \rightarrow \mathcal{H}_t,$$

a domain-only MKUFT candidate would require

$$\mathcal{D}(H_t) = \mathfrak{D}_P(\Xi_t)$$

with the resulting domain genuinely defining an appropriate Hamiltonian realisation.

More generally, if an independently supported relational variable is claimed to change the Hamiltonian/generator itself, that belongs in $\mathcal{T}_{P,t}$ and must be tested as such rather than being disguised as domain selection.

For a self-adjoint Hamiltonian,

$$U_t(\Delta) = \exp\left[-\frac{i}{\hbar} H_t \Delta\right],$$

$$\rho_{t+\Delta}^- = U_t(\Delta) \rho_t U_t(\Delta)^\dagger.$$

A quantum instrument $\{\mathcal{I}_o\}$ then supplies outcome probability and post-measurement physical state:

$$P(o) = \text{Tr} \mathcal{I}_o(\rho_{t+\Delta}^-),$$

$$\rho_{t+\Delta, o}^+ = \frac{\mathcal{I}_o(\rho_{t+\Delta}^-)}{P(o)}$$

when $P(o) > 0$. $\mathcal{O}$ stores/registers outcome o ; the physical state change remains P .

The capstone does not replace the Born rule. An independent I→P candidate must add predictive/interventional content beyond the strongest adequate P-state/history/boundary account and preserve no-signalling and established quantum limits.

13. General-relativistic specialisation

Classical general relativity supplies another structural comparator: admissible initial data obey Hamiltonian/momentum constraints, and well-defined Hamiltonian treatment of bounded regions depends on appropriate boundary terms and conditions.

Represent schematically

$$\mathfrak{L}_{G,t} = \left(\mathcal{D}_{G,t}^{\text{constraints}}, \mathcal{T}_{G,t}^{\text{Einstein/ADM}}, \mathcal{W}_{G,t} \right),$$

where $\mathcal{W}_{G,t}$ need not be present in a deterministic classical treatment.

This demonstrates that constraint surface before evolution is legitimate inside established gravity. It does **not** derive GR from MKUFT. The stronger burden is to derive why the effective GR law object should emerge from the deeper addressed architecture and recover observed gravity within stated tolerance.

14. I→P independent-content gate

The capstone gives a hierarchy:

Level 0 — P-only: the complete law object is determined by adequate P-state, history, boundary/environment and measurement context; I_t is recoverable from those variables. No independent I→P dynamics are established.

Level 1 — relational compression: I_t predicts or compresses the law object efficiently but is fully recoverable from P.

Level 2 — P-realised load-bearing relation: a relation intervention changes $\mathcal{D}$, $\mathcal{T}$ or $\mathcal{W}$, but an adequate P-description also predicts the change.

Level 3 — independent I candidate: after the strongest adequate P-state/history/boundary/measurement-context model is controlled, a separately specified I_t adds held-out or interventional prediction of

$$\mathcal{D}_{P,t}, \quad \mathcal{T}_{P,t}, \quad \text{or} \quad \mathcal{W}_{P,t}.$$

Only Level 3 begins to support independent I→P dynamics, and it must still preserve conservation laws, no-signalling and ordinary physical limits.

15. Recovery and falsification

The ordinary limit is mandatory. If relational/admissibility structure adds no independently measurable content beyond the adequate P-model, then

$$\mathfrak{L}_{P,t}^{\text{MK}} = \mathfrak{L}_{P,t}^{\text{standard}}$$

within the declared regime/tolerance, and therefore

$$\text{Propagate}_P^{\text{MK}} = \text{Propagate}_P^{\text{standard}}.$$

Reduce or reject the relevant branch if:

1. the active law object cannot be operationally separated from an adequate P-only description;
2. a proposed I deformation does not prospectively change the predicted domain, transition rule or weighting;
3. a simpler standard physical model predicts equally well;
4. path-history effects vanish under drift, hysteresis, instrument-memory and ordinary field-history controls;
5. scale promotion adds no intervention, capability or predictive gain over coarse-graining;
6. a quantum specialisation violates no-signalling, positivity, unitarity/complete positivity where required, or fails the standard quantum limit;
7. a gravitational specialisation cannot recover GR or an experimentally equivalent regime;
8. conservation fails without a measured exchange with boundary/environment or a new empirically supported law;
9. changed construction addresses are compared without a lawful map;
10. the capstone requires one untyped scalar or metric across incompatible S, I, P and O objects;
11. O is used to create missing P-dynamics without a physical/independently tested coupling;

12. an empty admissible domain is rescued by inventing an outcome rather than revising the model or declaring non-viability.

16. Prior art and contribution boundary

The capstone does not claim invention of constrained dynamics, viability theory, conditional/path measures, Markov kernels/categories, Maximum Caliber, Feynman–Kac weighting, self-adjoint extensions, quantum boundary conditions, quantum instruments, ADM constraints, fibers, quotients, coarse-graining, transport or holonomy.

Relevant prior/adjacent work includes:

- Fritz, T. (2020). *A synthetic approach to Markov kernels, conditional independence and theorems on sufficient statistics*. **Advances in Mathematics** 370, 107239. DOI [10.1016/j.aim.2020.107239](https://doi.org/10.1016/j.aim.2020.107239).
- Dixit, P. D. et al. (2018). *Perspective: Maximum caliber is a general variational principle for dynamical systems*. **Journal of Chemical Physics** 148, 010901. DOI [10.1063/1.5012990](https://doi.org/10.1063/1.5012990).
- Facchi, P., Garnero, G., and Ligabò, M. (2018). *Self-adjoint extensions and unitary operators on the boundary*. **Letters in Mathematical Physics** 108, 195–212. DOI [10.1007/s11005-017-1001-8](https://doi.org/10.1007/s11005-017-1001-8).
- Asorey, M., Clemente-Gallardo, J., and Muñoz-Castañeda, J. M. (2007/2008). *Boundary conditions: The path integral approach*. arXiv:0712.4353.
- Ozawa, M. (2001). *Quantum Measurement, Information, and Completely Positive Maps*. arXiv:quant-ph/0107090.
- Arnowitt, R., Deser, S., and Misner, C. W. (1962; arXiv repost 2004). *The Dynamics of General Relativity*. arXiv:gr-qc/0405109.
- Anco, S. C., and Tung, R. S. (2001). *Covariant Hamiltonian boundary conditions in General Relativity for spatially bounded spacetime regions*. arXiv:gr-qc/0109013.

The candidate MKUFT contribution is narrower:

factor the complete addressed S–I–P–O state into a typed admissibility descriptor, assemble a domain/transition/weight physical law object, propagate under that law, pass the result through a correctly typed $P \rightarrow (P, O)$ instrument, and recursively readdress the next effective possibility state; then connect that factorisation to the existing completion fibers, macro realisation classes, path-history transport, scale promotion, P-only null and changing-address guards.

Historical priority for that exact conjunction is not asserted without broader review.

17. Closure statement

The live-canon Update_SIP0 is no longer only an unnamed arrow.

The architectural chain is

$$\mathcal{U}_t + E_t \xrightarrow{\mathcal{Q}_t} \Xi_t \xrightarrow{\mathfrak{C}_P} \mathfrak{L}_{P,t} \xrightarrow{\text{Propagate}_P} P_{t+\Delta}^- \xrightarrow{\text{Instrument}_{P \rightarrow (P, O)}} (P_{t+\Delta}, O_{t+\Delta}) \xrightarrow{\text{Readdress}} \mathcal{U}_{t+\Delta}.$$

with

$$\mathfrak{L}_{P,t} = (\mathcal{D}_{P,t}, \mathcal{T}_{P,t}, \mathcal{W}_{P,t}).$$

Core rule:

the complete addressed state constrains admissibility; admissibility selects the physical law object appropriate to that state; physical dynamics run under that law; measurement remains a physical interaction plus a registered record; and the realised path becomes part of the address of what can happen next.

This is the current MKUFT drivetrain.

What remains after this capstone is not a missing architectural arrow. It is the branch-specific mathematical-physics programme: instantiate $\mathfrak{C}_P$ and $\mathfrak{L}_P$ for concrete regimes, derive accepted quantum/gravitational limits, and test whether any independently specified I-variable changes the law object beyond the strongest physical account.